

Francesco Congiu

PHD STUDENT

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Work Experience

University of Cagliari

Cagliari, Italy

TEACHING ASSISTANT

Mar 2026 - Present

- Developing Python hands-on notebooks and leading live-coding sessions on Data Structures, Algorithms (DP, Greedy, Graphs), and String Matching.
- Mentoring students on algorithmic efficiency (Big O) and Pythonic coding standards through LeetCode-style challenges and code reviews.

X-Ference

Cagliari, Italy

APPLIED SCIENTIST INTERN

Mar 2025 - Jul 2025

- Engineered a Confidentiality-Preserving RAG system using Python and HuggingFace, ensuring data privacy for enterprise-level document retrieval.
- Optimized retrieval accuracy by integrating custom EM-based reranking algorithms, reducing false-positive rates in query responses.
- Collaborated with the engineering team to deploy scalable LLM pipelines, improving internal document processing speed.

Adiacent

Cagliari, Italy

FULL STACK DEVELOPER INTERN

Mar 2023 - Jul 2023

- Architected a document digitization pipeline using C# and .NET, automating the ingestion of legacy records into modern databases.
- Designed a core infrastructure module currently utilized to manage 5.000+ digital assets.

Education

University of Cagliari

Cagliari, Italy

PH.D. IN TEACHING & LEARNING SCIENCES — SPECIALIZATION IN AI & EDUCATIONAL TECHNOLOGIES

Nov 2025 - Current

- **Research Areas:** Retrieval-Augmented Generation Systems, Confidentiality-Aware NLP, Information Retrieval.

University of Cagliari

Cagliari, Italy

M.S. IN COMPUTER SCIENCE: APPLIED ARTIFICIAL INTELLIGENCE, SUMMA CUM LAUDE

Sep 2023 - Sep 2025

- **Thesis:** "RetrievEM: Confidentiality-Preserving RAG via Expectation-Maximization"
- **Relevant Coursework:** Information Retrieval, Natural Language Processing, Machine Learning, Deep Learning, Adversarial and Reinforcement Learning.

University of Cagliari

Cagliari, Italy

B.S. IN COMPUTER SCIENCE

Sep 2020 - Jul 2023

- **Thesis:** "Development of a web application for blood donation management".
- **Relevant Coursework:** Algorithms & Data Structures, Software Engineering, Programming 1, Programming 2, Operating Systems

Publications

Confidentiality-Preserving RAG over Educational Resources

Cagliari, Italy

SIREM '25

Nov. 2025

- **Authors:** Boratto L., Congiu F., Fenu G., Marras M.
- Validated confidentiality-preserving architectures by benchmarking against Beir/FiQA dataset, achieving a MAP@10 of 0.3506 through an optimized Linear Fusion strategy.

Confidential Retrieval-Augmented Generation in Educational Contexts

Cagliari, Italy

2ND WORKSHOP ON ARTIFICIAL INTELLIGENCE WITH AND FOR LEARNING SCIENCES (WAILS '25)

Oct. 2025

- **Authors:** Boratto L., Congiu F., Fenu G., Medda G., Pau A.
- Introduced *Backfill*, a search-depth expansion algorithm improving HIT@10 by 0.09 on the Beir/FiQA benchmark via Linear Fusion.

Research Projects

RetrievEM: Open-source Framework for Confidentiality-Aware RAG

GitHub

CREATOR & LEAD MAINTAINER

Sep. 2025

- Designed and open-sourced a modular RAG framework integrating dense retrieval, cross-encoder reranking, and score-level fusion.
- Implemented an Expectation-Maximization (EM) inspired reranking algorithm to maximize query relevance while enforcing strict 100% confidentiality constraints.
- Developed a synthetic persona generation module to enable access-aware evaluations on benchmarks lacking native confidentiality metadata.